

EA Credit: Optimize Energy Performance

This prerequisite applies to

- ▶ BD+C: New Construction (1-18 points)
- ▶ BD+C: Core & Shell (1-18 points)
- ▶ BD+C: Schools (1-16 points)
- ▶ BD+C: Retail (1-18 points)
- ▶ BD+C: Data Centers (1-18 points)
- ▶ BD+C: Warehouses & Distribution Centers (1-18 points)
- ▶ BD+C: Hospitality (1-18 points)
- ▶ BD+C: Healthcare (1-20 points)

INTENT

To achieve increasing levels of energy performance beyond the prerequisite standard to reduce environmental and economic harms associated with excessive energy use and greenhouse gas emissions that disproportionately impact frontline communities.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Analyze efficiency measures during the design process and account for the results in design decision making. Use energy simulation of efficiency opportunities, past energy simulation analyses for similar buildings, or published data (e.g., Advanced Energy Design Guides) from analyses for similar buildings.

Analyze efficiency measures, focusing on load reduction and HVAC-related strategies (passive measures are acceptable) appropriate for the facility. Project potential energy savings and holistic project cost implications related to all affected systems.

Choose one of the options below.

Option 1. Energy Performance Compliance (1-18 points except Schools and Healthcare, 1-16 points Schools, 1-20 points Healthcare)

Demonstrate a Performance Cost Index (PCI)¹ below the Performance Cost Index Target (PCI_t) calculated in accordance with Section 4.2.1.1 of ANSI/ASHRAE/IESNA Standard 90.1-2016, Appendix G, Table 4.2.1.1. For mixed use buildings, the required PCI shall be calculated by using an area weighted average of the building types.

Calculate the PCI, PCI_t, and percentage improvement using metrics of cost and greenhouse gas (GHG) emissions.

Total LEED points for EAc Optimize Energy Performance are determined by summing the points documented in Table 1 (energy cost metric) plus the points documented in Table 2 (GHG emissions metric), where points for each Table are determined based on the project percent improvement PCI below the PCI_t documented for each metric.

Table 1. Points for percentage improvement in energy performance – Percent Cost PCI below PCI_t (1-9 points NC and CS, 1-8 points Schools, 1-10 points Healthcare)

New Construction	Healthcare, Major Renovation, CS	Points BD+C (except Schools, Healthcare)*	Points Healthcare	Points Schools
5%	2%	1	1	1
10%	5%	2	2	2
15%	10%	3	3	3
20%	15%	4	4	4
25%	20%	5	5	5
30%	25%	6	6	6
35%	30%	7	7	7
40%	35%	8	8	
45%	40%	9	9	8
50%	45%	EP	10	EP
	50%		EP	

Table 2. Points for percentage improvement in energy performance – Percent Greenhouse Gas Emissions PCI below PCI_t (1-9 points NC, 1-8 points Schools, 1-10 points Healthcare)

New Construction	Healthcare, Major Renovation, CS,	Points BD+C (except Schools, Healthcare)*	Points Healthcare	Points Schools
5%	2%	1	1	1
10%	5%	2	2	2
16%	10%	3	3	3
24%	16%	4	4	4
32%	24%	5	5	5
40%	32%	6	6	6
50%	40%	7	7	7
65%	50%	8	8	
80%	65%	9	9	8
100%	80%	EP	10	EP
			EP	

Table 1 and Table 2 Notes:

1. BD+C projects except Data Centers with Proposed Building Unregulated Energy exceeding 50% of the total Proposed Building Performance for the referenced metric (cost or GHG emissions), and BD+C: Data Centers projects with at least 40% gross colocation data center area may use the “Healthcare, Major Renovation, CS” column in lieu of the “New Construction” column.
2. Renewable Energy:
 - On-site renewable energy: For Table 1 and Table 2, on-site or on-campus renewable energy that meets ASHRAE Standard 90.1-2016 Section G 2.4.1 requirements for on-site renewable may be subtracted from the Proposed Building Performance in accordance with ASHRAE Standard 90.1-2016 Section G 2.4.1.
 - For the Table 2 greenhouse gas emissions metric, Tier 2 New off-site renewables documented in EA Credit Renewable Energy may be subtracted from proposed greenhouse gas emissions for electricity prior to calculating proposed building performance. Projects in locations where hourly electric greenhouse gas emissions factors are available must document performance for the baseline, proposed, and renewable generation using hourly electric emissions factors.

DATA CENTERS ONLY

In addition to the requirements above, analyze efficiency measures focused on IT load reduction and HVAC-related strategies. Collocated data centers may use the CS percentage improvement thresholds in lieu of NC thresholds to earn points in Table 1 and Table 2.

OR

Option 2. Prescriptive Compliance: ASHRAE Advanced Energy Design Guide (1-6 points, 1-4 points CS)

To be eligible for Option 2, projects must meet the Scope requirements of the applicable AEDGs (or combination of AEDGs for mixed use), and projects must use the ASHRAE 90.1-2016 Prescriptive compliance path in EA Prerequisite Minimum Energy Performance.

Implement and document compliance with the applicable recommendations and standards in Chapter 4, Design Strategies and Recommendations by Climate Zone, for the appropriate ASHRAE 50% Advanced Energy Design Guide and climate zone.

ASHRAE 50% Advanced Energy Design Guide for Small to Medium Office Buildings

- ▶ *Building envelope*: roofs, walls, floors, slabs, doors, continuous air barriers, and vertical fenestration (1 point, 2 points CS)
- ▶ *Interior and exterior lighting*, including daylighting and interior finishes (1 point).
- ▶ *Plug loads*, including equipment and controls (2 points, 0 points CS)
- ▶ *HVAC Systems and Controls* (2 points, 1 point CS).

ASHRAE 50% Advanced Energy Design Guide for Medium to Large Box Retail Buildings

- ▶ *Building envelope*: roofs, walls, floors, slabs, doors, vestibules, and fenestration – all orientations (1 point, 2 points CS)
- ▶ *Interior and exterior lighting*, excluding lighting power density for sales floor (1 point).
- ▶ *Additional interior lighting* for sales floor (1 point, 0 points CS)
- ▶ *Plug loads*, including equipment choices and controls (1 point, 0 points CS)
- ▶ *HVAC efficiency and control requirements* (2 points, 1 point CS).

ASHRAE 50% Advanced Energy Design Guide for K-12 School Buildings

(Not applicable for CS)

- ▶ *Building envelope*: roofs, walls, floors, slabs, doors, and vertical fenestration (1 point)
- ▶ *Interior and exterior lighting*, including daylighting and interior finishes (1 point)
- ▶ *Plug loads*, including equipment choices, controls, and kitchen equipment (2 points)
- ▶ *HVAC efficiency and control requirements* (2 points)

ASHRAE 50% Advanced Energy Design Guide for Large Hospitals

- ▶ *Building envelope:* roofs, walls, floors, slabs, doors, vestibules, continuous air barriers, and vertical fenestration (1 point, 2 points CS)
- ▶ *Interior and exterior lighting,* including daylighting (form or nonform driven) and interior finishes (1 point).
- ▶ *Plug loads,* including equipment choices, controls, and kitchen equipment (1 point, 0 points CS)
- ▶ *HVAC and Service Water Heating Systems and Equipment (2 points) (1 point CS.)*

ASHRAE 50% Advanced Energy Design Guide for Grocery Stores

- ▶ *Building envelope:* roofs, walls, floors, slabs, doors, vestibules, continuous air barriers, and vertical fenestration (1 point) (2 points – CS)
- ▶ *Interior and exterior lighting, including sales floor (1 points)*
- ▶ *Refrigeration, Plug, and Process loads,* including equipment choices and controls (2 points) (0 points CS)
- ▶ *HVAC efficiency and control requirements (1 point) (1 point CS.)*

Option 3. Systems Optimization (1-6 points)

To be eligible for Option 3, projects must use the ASHRAE 90.1-2016 Prescriptive compliance path in EA Prerequisite Minimum Energy Performance, and must not have more than 2,000 square feet of data center space, laboratory space, or manufacturing space.

Demonstrate an improvement beyond ASHRAE/ASHRAE/IESNA Standard 90.1-2016, with errata, for the following systems as applicable to the project scope: Interior and Exterior Lighting; Daylight controls; Building envelope; HVAC and service water heating equipment efficiency; and Equipment and appliances.

Use any combination of the strategies in any or all of the categories below, for a maximum of up to 6 points.

- ▶ *Interior and Exterior Lighting:*
 - 15% lighting power reduction (1 point)
 - 30% lighting power reduction (1 point)
 - 45% lighting power reduction (1 point)
- ▶ *Daylight controls:*

Install daylight-responsive controls for a given percentage of connected lighting load (lighting in non-regularly occupied space with occupant sensor controls may be excluded from connected lighting load).

 - 40% (1 point)
- ▶ *Building envelope:*
 - Climate Zones 1 – 2: Achieve 2 of the 3 strategies below for 1 point.
 - Thermal Mass Enclosure: More than 70% of opaque above-grade wall area meets ASHRAE 90.1-2016 definition for “mass wall”; and more than 70% of floor area meets ASHRAE 90.1-2016 definition for “mass floor”
 - 25% Envelope UA reduction
 - 50% SHGC reduction (including window shade factors)
 - Climate Zones 3 – 8: Achieve 2 of the 3 strategies below for 1 point.
 - 25% Envelope UA reduction
 - 50% Envelope UA reduction
 - 25% reduction in air infiltration measured during commissioning
- ▶ *HVAC and Service Water Heating Equipment Efficiency:*
 - Reduction in total fan power allowance of:
 - 20% (1 point)
 - Improvement in efficiency for at least 75% of the combined cooling, heating, and service water heating capacity
 - 20% (1 point)

- *Electric resistance heating except heat pump auxiliary heat must be included in total capacity.*
- ▶ **Equipment and Appliances:**
Install a percentage (by rated power) of eligible equipment and appliances meeting the following requirements:
 - *ENERGY STAR* equipment including appliances, office equipment, electronics, and commercial food service equipment (lighting and building envelope products are excluded from this credit). (Electronic Product Environmental Assessment Tool (EPEAT) equipment may be used in lieu of Energy Star equipment where applicable).
 - Prescriptive commercial kitchen and refrigeration equipment requirements listed in Appendix 3, Table 1.

The project scope of work must include at least 0.25 Watts per square foot of eligible equipment to apply this strategy.

Percent of Eligible Equipment Installed by Rated Power:

- ▶ 75% (1 point)

OR

Option 4. Data Centers only- System Optimization (1-3 points)

Calculate an Overall Systems Design Value as the sum of the maximum *design Mechanical Load Component (MLC)* and maximum *design Electrical Load Component (ELC)* in accordance with ASHRAE 90.4-2016 Section 6.2, Section 8.2, and Section 11. Document that the Overall Systems Design value is less than the Maximum Overall Systems Value by:

- 10% (1 point)
- 20% (2 points)
- 30% (3 points)

**If the electrical system design is incomplete, the design values shall be assumed to match the values in Table 8.2.1.1 and 8.2.1.2.*

Collocated data centers: document that the Overall Systems Design value is less than the Maximum Overall Systems Value by:

- 6% (1 point)
- 12% (2 points)
- 18% (3 points)

GUIDANCE

Refer to the consolidated guidance within EA Prerequisite Minimum Energy Performance, since the requirements and LEED documentation are closely linked for EA Prerequisite Minimum Energy Performance and EA Credit Optimize Energy Performance:

Behind the Intent

See EA Prerequisite Minimum Energy Performance, *Behind the Intent*.

Beta Update

Updated referenced standards and a new greenhouse gas emissions metric ensure that LEED continues to be a global leadership standard for energy performance and encourages owners to directly consider and address building carbon emissions. Revised prescriptive options to expand applicability to a

broader set of projects. Require design phase analysis for all Optimize Energy Performance options, including prescriptive and performance options.

Step-by-Step Guidance

See EA Prerequisite Minimum Energy Performance.

Assure that the additional Optimize Energy Performance requirement credit requirements are met:

- Analyze efficiency measures during the design process, focusing on load reduction and HVAC-related strategies (including passive measures) appropriate for the facility, project potential energy savings and holistic project cost implications, and account for the results in design decision making.

These requirements apply regardless of whether the project is using the performance path (Option 1) or prescriptive path (Options 2, 3 or 4) to document credit compliance.

The best approach for analyzing efficiency measures is a preliminary energy model, which evaluates heating and cooling load reduction strategies, passive HVAC strategies, and HVAC efficiency and control strategies (see EA prerequisite Minimum Energy Performance, *Further Explanation, Developing a Preliminary Energy Model* and *Further Explanation, Modeling HVAC Systems*). ASHRAE Standard 209 provides a standardized methodology which may be used for developing preliminary energy models that are used to inform the design process (See ASHRAE 209 Sections 6.3 - Load Reduction Modeling and 6.4 - HVAC System Selection Modeling).

Projects may also opt to use the AEDGs where applicable, and/or use modeling analyses previously performed for similar projects to meet the design phase analysis requirements.

Further Explanation

Calculations

See EA Prerequisite Minimum Energy Performance, *Further Explanation, Calculations* for calculation of the Performance Cost Index (PCI) and Performance Cost Index Target (PCI_t) using units of cost and GHG emissions.

Applying Renewable Energy Savings

See EA Prerequisite Minimum Energy Performance, *Further Explanation, Applying Renewable Savings* Only projects pursuing Option 1 of this credit may count savings from renewable energy systems.

Rating System Variations

Data Centers

See EA prerequisite Minimum Energy Performance, *Further Explanation, Rating System Variations*.

Co-located data centers may use the percentage improvement thresholds for Healthcare, Major Renovation, and CS in Table 1. Points for percentage improvement in energy performance - % Cost PCI below PCI_t and in Table 2. Points for percentage improvement in energy performance - % Greenhouse gas emissions PCI below PCI_t to determine points achieved under this credit.

Referenced Standards

- ▶ ASHRAE 90.1-2016
- ▶ ASHRAE 90.4-2016
- ▶ ASHRAE 50% Advanced Energy Design Guides
- ▶ ANSI/ASHRAE/IES Standard 90.1-2016 Performance Rating Method Reference Manual, PNNL 2017
- ▶ ASHRAE 209-2018
- ▶ Developing Performance Cost Index Targets for ASHRAE Standard 90.1 Appendix G - Performance Rating Method

Exemplary Performance

Option 1.

- ▶ Achieve 50% savings in the cost metric (45% for CS, Major Renovation, and Multifamily Residential), or
- ▶ Achieve 100% savings in the greenhouse gas emissions metric, (80% for CS, Major Renovation, and Multifamily Residential), or
- ▶ Use actual utility rates including all applicable seasonal, TOU, and demand components when calculating cost savings AND hourly greenhouse gas emissions factors when calculating greenhouse gas emissions savings.

Required Documentation

In addition to the documentation required for EA prerequisite Minimum Energy Performance, the following documentation is required for credit compliance

Documentation	Option 1	Option 2	Option 3	Option 4
Confirmation that energy efficiency measures were analyzed during the design phase and that the results were considered in design decision-making.	X	X	X	X
Target Finder results and summary	X (for applicable building types)			
Greenhouse gas emissions calculations that include credit for new off-site renewable energy (as applicable)	X (where applicable)			
AEDG compliance tables		X		
Calculations demonstrating achievement of systems optimization requirements (e.g. ASHRAE 90.1 lighting power density calculations, lighting fixture calculations demonstration portion of lighting power with daylighting control, etc.)			X	
ASHRAE 90.4 Calculations (consistent with documentation requirements listed in ASHRAE 90.4)				X

Connection to Ongoing Building Performance

- ▶ LEED O+M EA credit Energy Performance: Designing for higher levels of energy efficiency helps decrease building operating costs and reduces the environmental and economic harms associated with excessive energy use throughout the building life cycle; this may help improve the building's energy performance score.